

METHODS FOR COMPARING TWO GROUPS

Some of them

SUMMARY OF THE METHODS

Two sample problems are among the most common in statistics.

We begin by sketching a summary of the methods available to compare two groups. Methods are classified attending to two criteria: the nature of the response variable Y (the variable being compared) and the kind of samples we have collected from the two groups (dependent or independent).

DEPENDENT SAMPLES AND INDEPENDENT SAMPLES

Dependent samples: observations in each group are meaningfully matched with one another.

- A study observes 15 nursing-home patients with dementia. The number of episodes of aggressive behavior was recorded each day for 12 weeks. A day is called a *moon day* if it is the day of a full moon and *other* otherwise. The average number of aggressive incidents for moon days and other days is compared.
- Researchers at a pharmaceutical company tested a new sunscreen formula. They applied the new formula to one arm and the old formula to the other arm of each of a set of individuals.
- Measures taken on twins or couples: one of them being in one sample and the other in the other sample (matched-pairs).

Two groups

Independent samples

- Y quantitative
 - t.test()*
 - Wilcoxon Rank-Sum test
- Y categorical
 - χ^2 test of homogeneity or independence

* Assumptions required

Dependent samples

- Y quantitative
 - t.test(..., paired =T, ...)*
 - Sign test
 - Wilcoxon signed-rank test
- Y categorical with two levels
 - McNemar's test

More than Two groups

Independent samples

- Y quantitative
 - One Way ANOVA*
 - Kruskal Wallis test
- Y categorical
 - χ^2 test of homogeneity or independence

* Assumptions required

Dependent samples

- Y quantitative
 - One Way ANOVA with repeated measures*
 - Friedman's rank sum test
- Y categorical with two levels
 - Cochran's Q test

OPTIONS WITH NON NORMAL DATA (I)

Options we can use when the population distribution is non Normal or we have only a few observations:

- Sometimes we can transform our data so that their distribution is more nearly Normal (see two slides forward).
- In some settings, other standard distributions replace the Normal distributions as models for the overall pattern in the population (Exponential, Weibull, Poisson, ...)
- Permutation tests cannot be used for populations that differ under the null hypothesis, for general relationships, or in situations where we cannot resample and preserve the study design.
- Bootstrap methods and permutation tests do not require Normality or any other specific form of the sampling distribution. You can use these methods unless the sample is so small that it may not represent the population well.

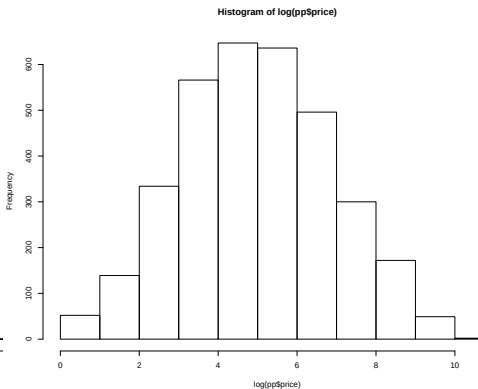
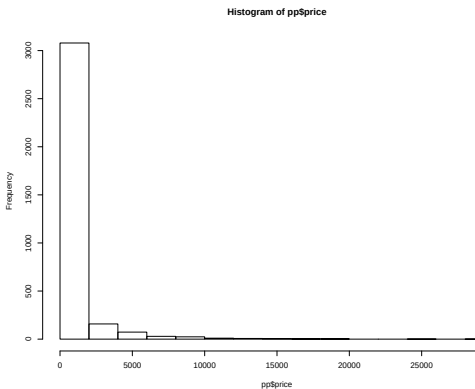
OPTIONS WITH NON NORMAL DATA (II)

Options we can use when the population distribution is non Normal or we have only a few observations:

- There are other **nonparametric** methods that do not require any specific form for the distribution of the population. Common nonparametric methods do not make use of the actual values of the observations. They use counts of observations or work with the rank (place in order) of each observation in the set of all data (rank tests). They are **robust to outliers**.

In these slides we'll talk about one nonparametric test based on counts (**the sign test**) and two rank tests (**Wilcoxon signed-rank test** and **Wilcoxon rank sum test**). We will also discuss the χ^2 **test of independence** and **McNemar's test for dependent samples (matched pairs)**.

TRANSFORMATIONS



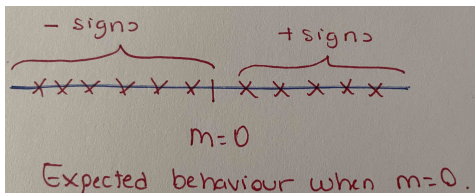
THE SIGN TEST

- Paired data, to test whether a treatment is effective.
- It uses the data on the differences between both treatments; it uses only the information provided by the sign of those differences. It ignores observations pairs for which the difference is 0.
- The only thing needed is to evaluate which observation in a matched pair has the better response.
- The null hypothesis is that the median of the differences is 0.

For instance, for each subject the only information recorded is which treatment has given the best result.

THE SIGN TEST. NULL HYPOTHESIS

The random variable we are going to use is the difference between the treatment values. We assume this variable has an unknown median m . Upon taking a random sample we are interested in testing whether the median is equal to 0, i.e., $H_0 : m = 0$.



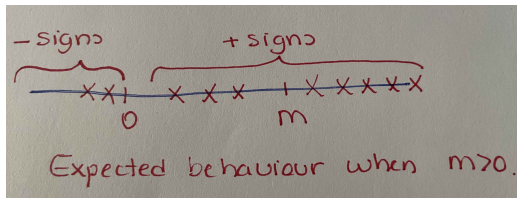
Under the null hypothesis the number of positive and negative differences, N_+ and N_- , both follow a binomial distribution with parameters n and $p = 0.5$. That is:

$$N_+ \sim B(n, 0.5) \quad N_- \sim B(n, 0.5)$$

THE SIGN TEST. UNILATERAL TEST $H_1 : m > 0$

Under $H_1 : m > 0$, we should expect more positive signs that would be expected if the null were true. In this case, we should reject the null if n_+ is too large:

$$p\text{-value} = P(N_+ \geq n_+) \quad N_+ \sim \text{Bin}(n, p = 0.5)$$



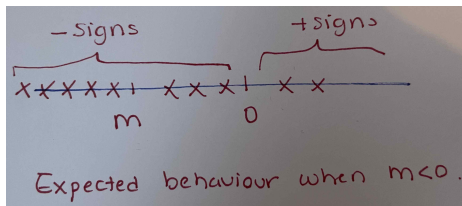
$$\left. \begin{array}{l} H_0 : m = 0 \\ H_1 : m > 0 \end{array} \right\} \Leftrightarrow \left. \begin{array}{l} H_0 : m \leq 0 \\ H_1 : m > 0 \end{array} \right\} \Leftrightarrow \left. \begin{array}{l} H_0 : p \leq 0.5 \\ H_1 : p > 0.5 \end{array} \right\}$$

where p is the proportion of positive values in the differences, i.e., $p = P(X > Y) = P(X - Y > 0)$ where X is the result of the first treatment and Y the result for the second one.

THE SIGN TEST. UNILATERAL TEST. $H_1 : m < 0$

Under $H_1 : m < 0$, we should expect more negative signs that would be expected if the null hypothesis were true. In that case, we should reject the null hypothesis if n_+ , the observed number of positive signs, is too small:

$$p\text{-value} = P(N_+ \leq n_+) \quad N_+ \sim \text{Bin}(n, p = 0.5)$$



The following three tests are equivalent:

$$\left. \begin{array}{l} H_0 : m = 0 \\ H_1 : m < 0 \end{array} \right\} \Leftrightarrow \left. \begin{array}{l} H_0 : m \geq 0 \\ H_1 : m < 0 \end{array} \right\} \Leftrightarrow \left. \begin{array}{l} H_0 : p \geq 0.5 \\ H_1 : p < 0.5 \end{array} \right\}$$

where p is the proportion of positive values in the differences, i.e., $p = P(X > Y) = P(X - Y > 0)$ where X is the result of the first treatment and Y the result for the second one.

THE SIGN TEST. BILATERAL TEST. $H_1 : m \neq 0$

When testing $H_0 : m = 0$ versus the alternative $H_1 : m \neq 0$ it makes sense that we should reject the null hypothesis if we have too few negative signs or too few positive signs. Formally, we reject if N_{min} , which is defined as the smaller of n_- and n_+ , is too small. The p-value is then:

$$p - value = 2P(N_{min} \leq \min(n_-, n_+))$$

THE SIGN TEST. EXAMPLE 1

In a crossover design to compare two different treatments to relieve headache, participants are randomly assign to one of them. After a reasonable time, they would follow the other treatment and record which one had a better result. In a sample of $n = 200$ subjects, 120 subjects found the second treatment was better to relieve headaches. Do these data provide evidence on the second treatment being better than the first one?

THE SIGN TEST. EXAMPLE 2

We want to compare two tanning methods without exposure to the sun to avoid skin cancer risk. We want to compare two treatments—a bronze tanning lotion applied twice over a two-day period, or a tanning studio where the person is exposed to ultraviolet (UV) light. We'll refer to these treatments as “tanning lotion” and “tanning studio.” In a crossover design, 5 participants get a tan using one method (randomly chosen) and after it wears off they get a tan using the other method. For each participant, we observe which treatment gives the better tan.

The hypothesis to test is that the tanning studio would give a better tan even though the lotion is cheaper.

The results of the experiment were that the tanning studio gave a better tan than the tanning lotion for four of the five participants.

THE SIGN TEST. EXAMPLE 3

A new medicine against hypertension was tested on 18 patients. After 40 days of treatment, the following changes (after – before) of the diastolic blood pressure were observed.

$-5, -1, +2, +8, -25, +1, +5, -12, -16$

$-9, -8, -18, -5, -22, +4, -21, -15, -11$

Use the sign test to determine if the medicine has the desired effect on blood pressure. What is the p-value?

THE WILCOXON SIGNED-RANK TEST

- Same type of data as before (paired data)
- For each matched pair of responses, it uses the difference between the responses. It ignores observations pairs for which the difference is 0.
- It uses the quantitative information provided by the n difference scores by ranking their magnitudes, in absolute value.
- The test statistic (let's call it W) is the sum of the ranks for the differences that are positive.
- It assumes that the differences of observations have a symmetric population distribution. The null hypothesis is that the population median of difference scores is 0.

THE WILCOXON SIGNED-RANK TEST

The mean and variance of the statistic are:

$$E(W) = \frac{n(n+1)}{4}, \quad \text{Var}(W) = \frac{n(n+1)(2n+1)}{24}, \text{ if } H_0 \text{ is true}$$

- For small n , the exact distribution of the test statistic can be computed (when there are not ties in the ranked data). In the presence of ties, software might show a warning.
- For large n , or when ties are present in the data, an asymptotic test using the Normal distribution to calculate the p-value can be performed.

THE WILCOXON SIGNED-RANK TEST. EXAMPLE 3 REVISITED

Data: 18 observed differences on blood pressure values after treatment - before treatment. The absolute value of the differences are ordered and ranks are assigned (in red). The mean rank is assigned when ties are present.

-1	+1	+2	+4	-5	-5	+5	-8	+8	-9
1.5	1.5	3	4	6	6	6	8.5	8.5	10
-11	-12	-15	-16	-18	-21	-22	-25		
11	12	13	14	15	16	17	18		

$$W = 1.5 + 3 + 4 + 6 + 8.5 = 23$$

There are ties in the data. We cannot compute an exact p-value. For the alternative $H_1 : m < 0$, we use the normal approximation:

$$p - \text{value} = P(W \leq 23)$$

THE WILCOXON SIGNED-RANK TEST. EXAMPLE 4

In a study, 5 food experts compared food made with hot oil and a new oil-free fryer. They rated the taste of hash browns prepared with each method.

	Expert				
	1	2	3	4	5
Hot oil	78	84	62	73	63
Oil free	75	85	67	75	66

Since there are ties in the differences between both methods, the p-value is computed using the normal approximation:

```
hot_oil=c(78,84,62,73,63)
oil_free=c(75,85,67,75,66)
wilcox.test(hot_oil- oil_free,exact=F)
```

SIGN TEST VS. THE WILCOXON SIGNED-RANK TEST

Which do most students spend more time doing, browsing the Internet or watching TV? Test they watch TV more.

Student	Internet	TV
1	60	120
2	20	120
$\vdots$	$\vdots$	$\vdots$
54	60	90

Sign test: It uses only the fact that 35 students spent more time watching TV and 19 students spent more time browsing the Internet.

Wilcox signed rank test:

```
wilcox.test(browse, WatchTV, paired=TRUE, exact=FALSE, alternative="less")
```

THE WILCOXON RANK SUM TEST

- We have two independent random samples of sizes n_1 and n_2 from two groups, obtained either from random sampling or a randomized experiment.
- The null hypothesis is that the two independent samples come from the same population.

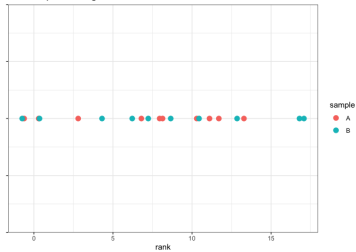
$$\left. \begin{array}{l} H_0 : F = G \quad \text{The two distributions are the same} \\ H_1 : F \neq G \quad \text{One distribution is systematically larger} \end{array} \right\}$$

- In order to be a test of medians (m), the distributions of values for each group need to be of similar shape and spread. Otherwise the test is typically a test of stochastic equality.

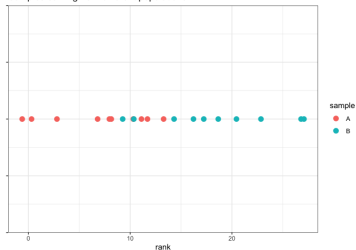
$$\left. \begin{array}{l} H_0 : m_1 = m_2 \\ H_1 : m_1 \neq m_2 \end{array} \right\}$$

THE WILCOXON RANK SUM TEST

Two samples coming from the same distribution



Samples coming from different populations



THE WILCOXON RANK SUM TEST

The procedure:

- Rank all $n = n_1 + n_2$ observations.
- The statistic W is **the sum of the ranks for the first sample**.
- Under the null hypothesis, its mean and standard deviation is:

$$\mu_W = \frac{n_1(n+1)}{2} \quad \sigma_W = \sqrt{\frac{n_1 n_2 (n+1)}{12}}$$

- The Wilcoxon rank sum test rejects the hypothesis that the two populations have identical distributions when the rank sum W is far from its mean.

THE WILCOXON RANK SUM TEST

Observation:

- The exact distribution of the Wilcoxon rank sum test is obtained assuming that all observations in both samples take different values. However, we often find observations tied at the same value. In this case, exact p-values cannot be computed. Use the Normal approximation instead.

How do ranks tests compare to permutation tests?

- Calculation of the sampling distribution under H_0 is simpler for rank tests because it depends only on the sizes of the samples.
- Compared to permutation tests, rank tests are recommended for very small sample sizes that are clearly non Normal and when exact p-values cannot be computed for permutation tests.

THE WILCOXON RANK SUM TEST. AN EXAMPLE

An experiment designed to compare two treatments against corrosion yielded the following data in pieces of wire subjected to the two treatments.

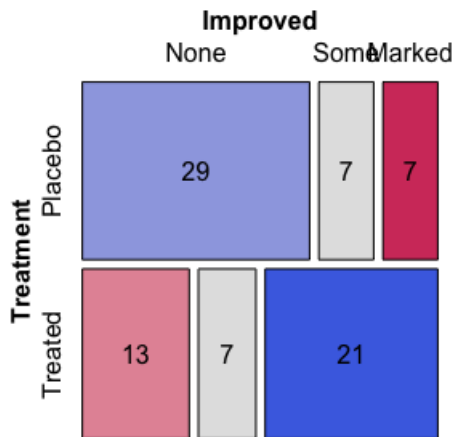
Treatment 1	65.2, 67.1, 69.4, 78.2, 74, 80.3
Treatment 2	59.4, 72.1, 68.0, 66.2, 58.5

The ordered values are 58.5, 59.4, 65.2*, 66.2, 67.1*, 68, 69.4*, 72.1, 74*, 78.2*, 80.3*.

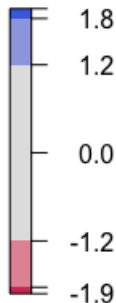
$$W = 3 + 5 + 7 + 9 + 10 + 11 = 45$$

The p-value, for an exact bilateral test, is 0.1255. The null hypothesis that both samples come from the same distribution cannot be rejected. We'll compute the p-value for a Normal approx. on the blackboard.

CONTINGENCY TABLES



Pearson residuals:



p-value =
0.002

CHI-SQUARED TEST

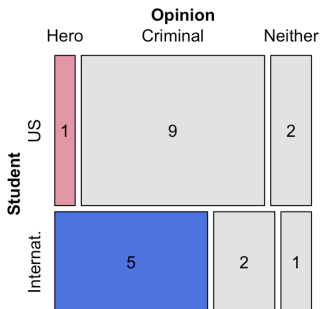
For the χ^2 test of independence to be valid:

- Expected counts in each cell ≥ 5 . Nowadays, most agree that we can have $E_j < 5$ for some of the cells (say, 20 % of them). Some of the E_j can be as small as 2, but none of them should fall below 1. If that happens use option `simulate.p.value=TRUE` in the function `chisq.test`.
- Conclusion: report p-value and interpret in context. If a decision is needed, reject H_0 when p-value < 0.05 . As with any hypothesis test *failing to reject the null hypothesis doesn't mean the variables are definitely independent. All that can be said is that independence is still plausible and can't be ruled out.*

CHI-SQUARED PERMUTATION TEST

Edward Snowden: Hero or Criminal?

	Student	Opinion	RanPerm
1	US	Hero	US
2	Internat.	Hero	US
3	Internat.	Hero	US
4	Internat.	Hero	Internat.
5	Internat.	Hero	US
6	Internat.	Hero	US
7	US	Criminal	US
8	US	Criminal	Internat.
9	US	Criminal	Internat.
10	US	Criminal	US
11	US	Criminal	US
12	US	Criminal	Internat.
13	US	Criminal	Internat.
14	US	Criminal	Internat.
15	US	Criminal	US
16	Internat.	Criminal	Internat.
17	Internat.	Criminal	US
18	US	Neither	US
19	US	Neither	US
20	Internat.	Neither	Internat.



Pearson residuals:
1.7
1.2
0.0
-1.4
p-value = 0.021

Permutation p-value for the chi-squared test 0.021.

LIMITATIONS OF CHI-SQUARED TEST

If the p-value is very small, strong evidence exists against the null hypothesis of independence. We can infer that the variables are associated.

The chi-squared statistic and the associated p-value tell us nothing about **the nature** or **the strength** of the association.

You may be tempted to regard a large value of the χ^2 statistic (or a very small p-value) in the test of independence as indicating a strong association. This is a **misuse** of the statistic.

DETERMINING THE STRENGTH OF THE ASSOCIATION

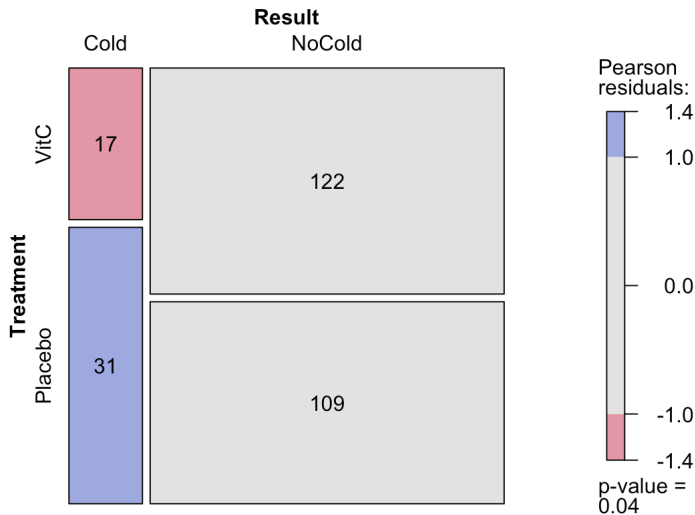
Three questions are normally addressed in analyzing contingency tables:

- Is there an association? The chi-squared test of independence addresses this. When the p-value is small, we infer that the variables are associated.
- How do the cell counts differ from what independence predicts? To answer this, interpret colors in the mosaic plot.
- How strong is the association? Analyzing the strength of the association reveals whether the association is an important one or if it is statistically significant but weak and unimportant in practice.

MEASURES ASSOCIATION (2×2 TABLES)

- Difference of proportions.
- Ratio of proportions: Relative Risk or risk ratio.
 - Popular measure for describing the association between a drug and a control treatment in medical studies.
- Odds and Odds ratio.
 - The odds in a given row are defined as the ratio of the two conditional proportions in that row.
 - The odds ratio is then the ratio of the two odds.

MEASURES OF ASSOCIATION (2×2 TABLES)

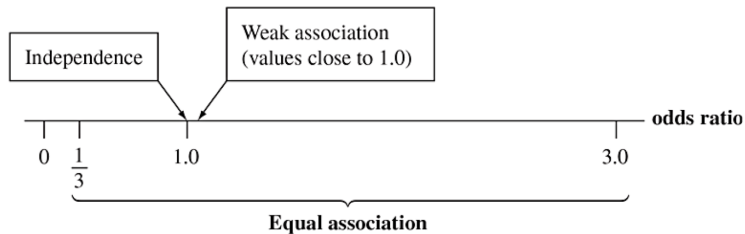


MEASURES OF ASSOCIATION (2×2 TABLES)

Properties of the odds ratio:

- The odds ratio can equal any nonnegative number.
- Under independence, the odds ratio equals 1.
- Values farther from 1.0 in either direction represent stronger associations. Two values for the odds ratio represent the same strength of associations in opposite direction, when one value is the reciprocal of the other.

ODDS RATIO INTERPRETATION (2×2 TABLES)



Values of the Odds Ratio farther from 1 represent stronger association.

INTERPRETATIONS (2×2 TABLES)

- Relative risk:
 - Those who take VitC. are 0.55 times more likely to get a cold than those taking a placebo. Those taking placebo are $1/0.55 = 1.81$ times more likely to get a cold than those taking VitC.
- Odds ratio:
 - The odds of getting a cold given VitC. are 0.49 times (half) the odds of getting a cold given placebo. The odds of getting a cold given placebo are $1/0.49 = 2.04$ times greater.

CHALLENGER EXPLOSION (1986)

Plot Source



A major malfunction

Challenger's brief flight

.578 seconds

Following Challenger's liftoff, a puff of black smoke — seen only by automatic launch cameras — indicates a problem with one of the O-ring seals at the joint between segments of the shuttle's right-hand solid rocket booster.

No human eyes see the smoke, and there would have been no way to abort the flight if they had.

58 seconds

A small jet of smoke and flame bursts through the side of the booster and quickly grows.

73 seconds

The flame burns through the strut attaching the solid rocket booster to the external fuel tank, causing the booster to swivel into the side of the tank. The resulting massive explosion destroys the space shuttle.

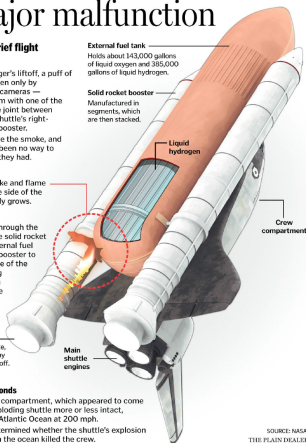
Full thrust

Once the booster's ignite, there is no way to shut them off.

3 minutes, 58 seconds

Challenger's crew compartment, which appeared to come away from the exploding shuttle more or less intact, smashes into the Atlantic Ocean at 200 mph.

Officials never determined whether the shuttle's explosion or the impact with the ocean killed the crew.



SOURCE: NASA
THE PLAIN DEALER

CHALLENGER EXPLOSION (1986)

Challenger launches on Jan 27, 1986 at $29^{\circ}F$. Are there more O-ring failures at lower temperatures?

Response variable: number of incidents $\{0, 1, 2, 3\}$

Explanatory variable: Cool weather or Warm weather

There is data from 24 launches previous to the Challenger.

	N of incidents			
Weather	0	1	2	3
Warm	17	2	1	0
Cold	0	3	0	1

The table is sparse and conditions for the χ^2 are hard to meet. With such sensitive data, let's use the Wilcoxon Rank Sum test.

CHALLENGER EXPLOSION (1986)

Incidents	Launch	order	rank
0	Warm	1	9
0	Warm	2	9
0	Warm	3	9
0	Warm	4	9
0	Warm	5	9
0	Warm	6	9
0	Warm	7	9
0	Warm	8	9
0	Warm	9	9
0	Warm	10	9
0	Warm	11	9
0	Warm	12	9
0	Warm	13	9
0	Warm	14	9
0	Warm	15	9
0	Warm	16	9
0	Warm	17	9
1	Cool	18	20
1	Cool	19	20
1	Cool	20	20
1	Warm	21	20
1	Warm	22	20
2	Warm	23	23
3	Cool	24	24

Ordered data with ranks

Compute the test statistic

Add ranks for the smallest group

$$W = 20 + 20 + 20 + 24 = 84$$

exact p-value $P(W \geq 84)$

CHALLENGER EXPLOSION (1986)

Elements of smaller group	No. of groupings	Value of W
0 0 0 0	2380	36
0 0 0 1	3400	47
0 0 0 2	680	50
0 0 1 1	1360	58
0 0 0 3	680	51
0 0 1 2	680	61
0 1 1 1	170	69
0 0 1 3	680	62
0 1 1 2	170	72
1 1 1 1	5	80
0 0 2 3	136	65
0 1 1 3	170	73
1 1 1 2	10	83
0 1 2 3	85	76
1 1 1 3	10	84
1 1 2 3	10	87

List all the ways of reassigning the observed incidents to the Cool group

Total number of ways $\binom{24}{4} = 10626$

Exact p-value

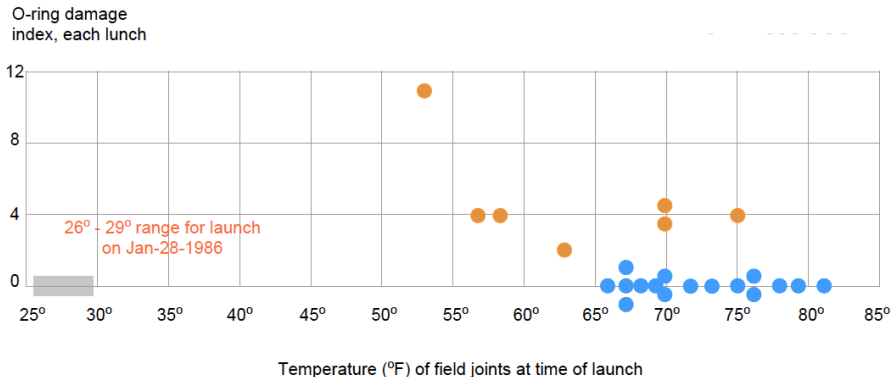
$$P(W \geq 84) = \frac{10+10}{10626} = 0.0018$$

There is compelling evidence that the number of accidents is systematically larger in Cold weather.

Solving it with R can be found in `examples_slides.R`

CHALLENGER EXPLOSION (1986)

24 launches prior to the Challenger



Data and Plot Source

LET'S RECAP A BIT

Choosing a statistical test

